

Determinants



2x2 determinants

- 1 Find the determinant of the matrix with rows $\langle 4, 3 \rangle, \langle 2, 5 \rangle$.
 - 2 Find the determinant of the matrix with rows $\langle -2, 1 \rangle, \langle 3, 4 \rangle$.
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3x3 determinants via cofactor expansion

- 3 Find the determinant of the matrix with rows $\langle 1, 2, 3 \rangle, \langle 0, 1, 4 \rangle, \langle 5, 6, 0 \rangle$ by cofactor expansion along the first row.
 - 4 Find the determinant of the matrix with rows $\langle 2, 0, 1 \rangle, \langle 3, 1, 0 \rangle, \langle 1, 2, 4 \rangle$ by cofactor expansion along the first row.
 - 5 Recompute the determinant of the matrix with rows $\langle 1, 2, 3 \rangle, \langle 0, 1, 4 \rangle, \langle 5, 6, 0 \rangle$ by expanding along the SECOND row instead, to confirm the answer matches.
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The determinant product rule

- 6 For A with rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ and B with rows $\langle 2, 1 \rangle, \langle 0, 3 \rangle$, verify $\det(AB) = \det(A)\det(B)$.
 - 7 Using the product rule $\det(AB) = \det(A)\det(B)$, explain why $\det(A^{-1}) = \frac{1}{\det A}$ for an invertible matrix A .
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Determinants and scalar multiples / row operations

- 8 If A is a 2×2 matrix with $\det A = 5$, find $\det(2A)$ (scaling EVERY entry of A by 2).
 - 9 If A is a 3×3 matrix with $\det A = 5$, find $\det(2A)$.
 - 10 Swapping two rows of a matrix negates its determinant. If A has rows $\langle 1, 2 \rangle, \langle 3, 4 \rangle$ with $\det A = -2$, find the determinant of the matrix with rows swapped: $\langle 3, 4 \rangle, \langle 1, 2 \rangle$.
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The invertibility criterion

- 11 State the invertibility criterion in terms of the determinant.

- 12 Is the matrix with rows $\langle 2, 4 \rangle, \langle 1, 2 \rangle$ invertible? Justify using the determinant.
- 13 Is the matrix with rows $\langle 1, 0, 2 \rangle, \langle 0, 3, 1 \rangle, \langle 2, 0, 4 \rangle$ invertible? Justify using the determinant.
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Applying the invertibility criterion

- 14 For what value(s) of k is the matrix with rows $\langle k, 2 \rangle, \langle 3, k \rangle$ NOT invertible?
- 15 For what value(s) of k is the matrix with rows $\langle 1, k \rangle, \langle k, 4 \rangle$ NOT invertible?
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Geometric meaning of the determinant

- 16 For a 2×2 matrix A with rows $\mathbf{r}_1, \mathbf{r}_2$, $|\det A|$ gives the area of the parallelogram spanned by $\mathbf{r}_1$ and $\mathbf{r}_2$. Find this area for $\mathbf{r}_1 = \langle 3, 0 \rangle, \mathbf{r}_2 = \langle 1, 4 \rangle$.
- 17 Explain, using the geometric meaning of the determinant, why a matrix with two identical rows always has determinant 0.
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Determinants of special matrices

- 18 Find the determinant of the identity matrix I_3 (rows $\langle 1, 0, 0 \rangle, \langle 0, 1, 0 \rangle, \langle 0, 0, 1 \rangle$).
- 19 Find the determinant of the matrix with rows $\langle 3, 0, 0 \rangle, \langle 0, -2, 0 \rangle, \langle 0, 0, 5 \rangle$ (a diagonal matrix).
- 20 Find the determinant of the matrix with rows $\langle 0, 0, 3 \rangle, \langle 0, 1, 0 \rangle, \langle 3, 0, 0 \rangle$, and explain why it is nonzero even though the main diagonal has two zeros.
- 21 Find the determinant of the zero matrix (all entries 0) of any size, and connect this to the invertibility criterion.
- 22 Explain why any matrix with an entire row (or column) of zeros must have determinant 0, using cofactor expansion along that row.
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Cramer's Rule

- 23 Cramer's Rule solves $ax + by = e, cx + dy = f$ via $x = \frac{\det A_x}{\det A}, y = \frac{\det A_y}{\det A}$, where A is the coefficient matrix and A_x, A_y replace the respective column with $\langle e, f \rangle$. Use it to solve $2x + 3y = 8, x - y = 1$.
- 24 Explain why Cramer's Rule fails (gives no unique solution) whenever $\det A = 0$.

- 25** Compare Cramer's Rule to Gaussian elimination as a solution method for large systems (say, 10 equations in 10 unknowns). Which is more practical, and why?